

Master Thesis

Enhancing Relational Understanding in CLIP
Leveraging Hard Negative Captions (HNC)

Wen Wen

Supervisor: Dr. Pascal Tilli
Prof. Dr. Ngoc Thang Vu

Outline

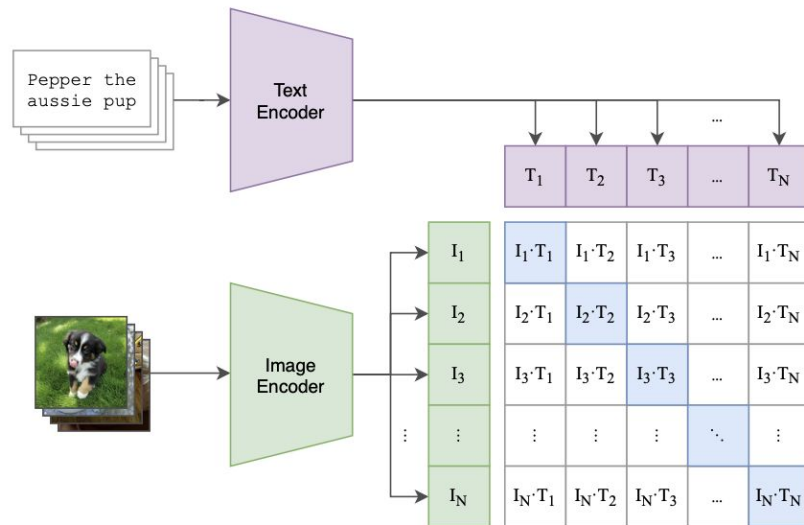
- Background Knowledge
- Research Question
- Method
- Experiments
- Results and Analysis
- Visual Interpretability
- Conclusions

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Background Knowledge

- **CLIP** (Contrastive Language–Image Pretraining)
- **Dual-encoder model**
- Using a **contrastive loss**
- Reward the correct image–caption pairing and penalize all other pairings in the batch
- Connect images and text in a shared embedding space
- **Global** embedding alignment
- Strong **Zero-shot** ability



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Research Question

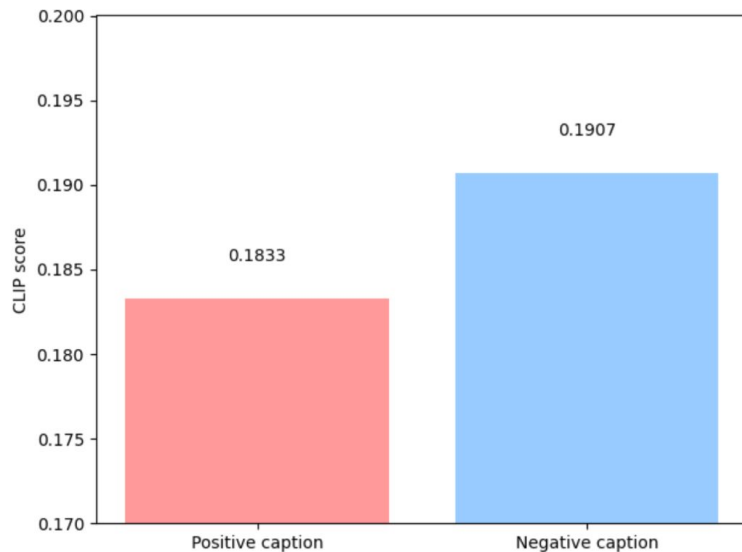
CILP cannot handle relations in images well.



The trees are **behind** the fence



The trees are **in front of** the fence



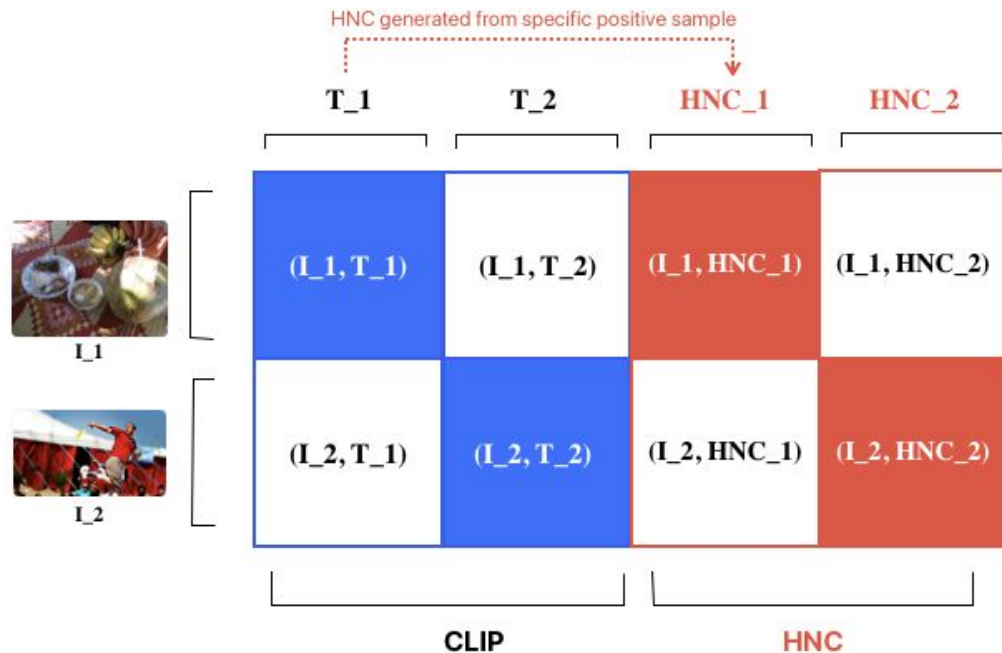
Goal: Improve relational sensitivity while preserving object recognition capabilities.

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Method:

→ Inject local information for global embedding alignment
By combine contrastive learning with HNC



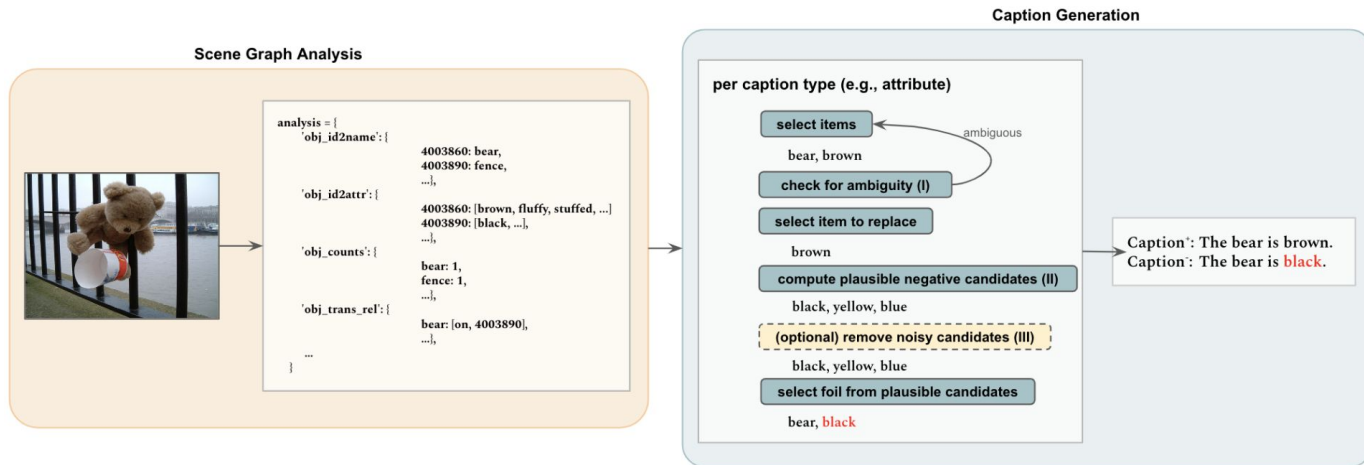
Blue part:
Standard Contrastive learning

Red Part:
Additional HNC part

Hard Negative Caption(HNC) Dataset

From GQA dataset

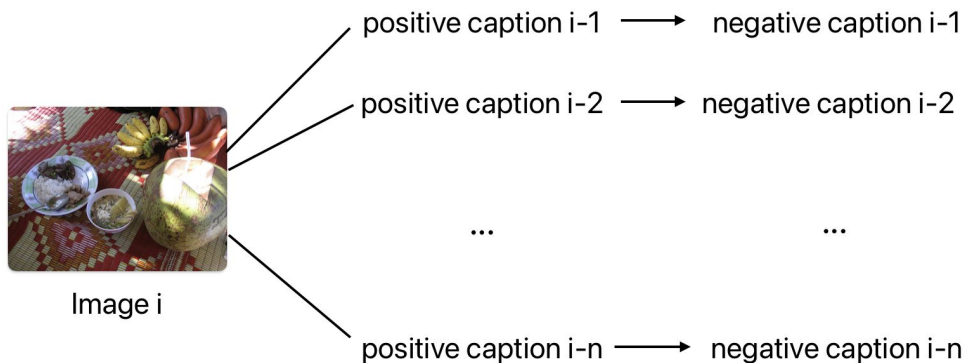
For each scene graph, this pipeline is run through to generate hard negative captions.



Thanks to Dr. Pascal Tilli and their team.

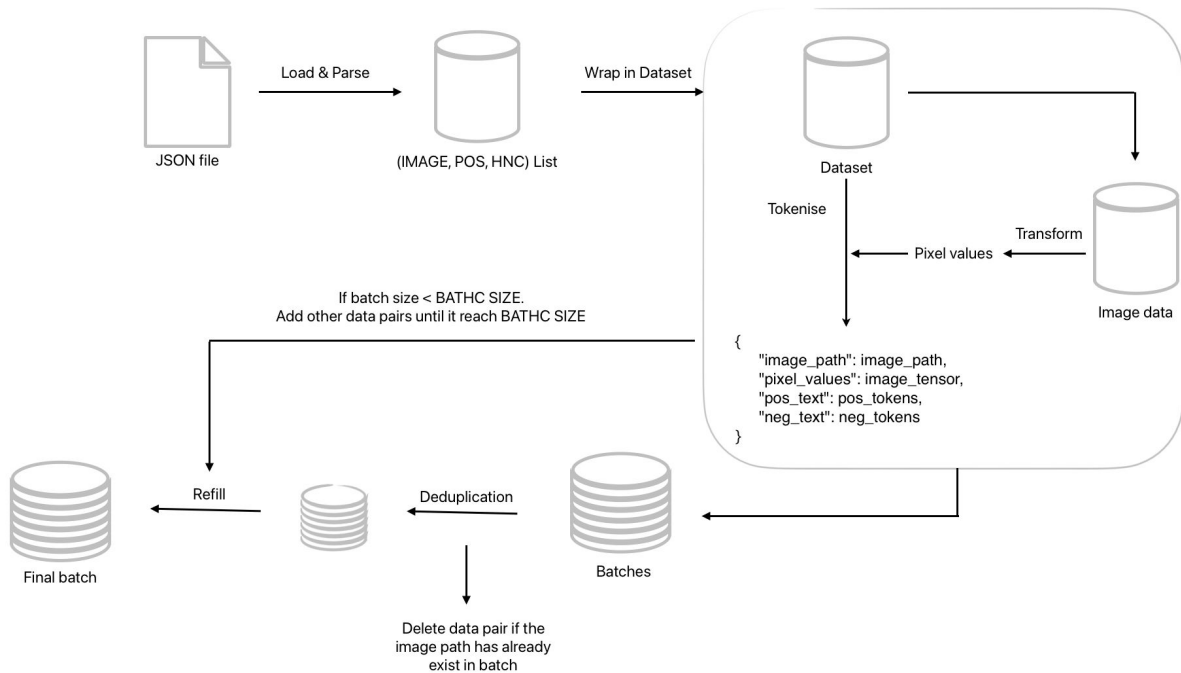
<https://aclanthology.org/2023.conll-1.24.pdf>

Data Preprocessing

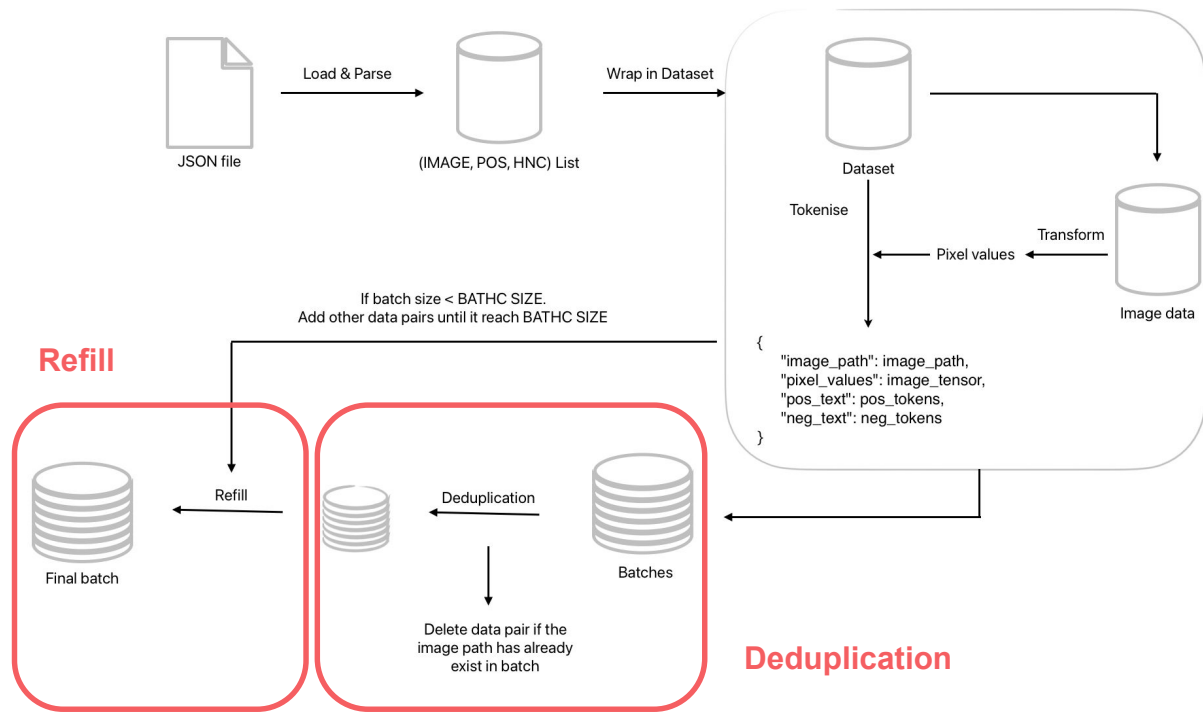


- One image has multiple positive captions and relative hard negative captions.
- For contrastive learning, it's important to remove the repeated image path in each batch.
- Otherwise, repeated positive captions can be treated as random negative captions for that image to influence training

Data Preprocessing

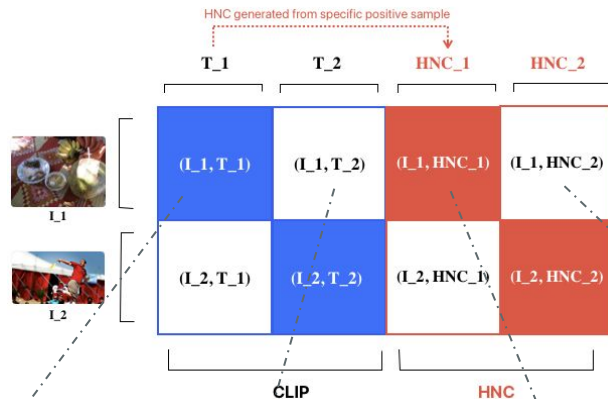


Data Preprocessing



To enforce image uniqueness within each training batch

Loss Function



$$\mathcal{L}_i = -\frac{1}{N} \sum_{i=1}^N \log \left(\exp \left(\frac{\text{sim}(I_i, T_i)}{\tau} \right) + \sum_{j \neq i} \exp \left(\frac{\text{sim}(I_i, T_j)}{\tau} \right) + \alpha \exp \left(\frac{\text{sim}(I_i, T_i^{\text{HNC}})}{\tau} \right) + \sum_{j \neq i} \exp \left(\frac{\text{sim}(I_i, T_j^{\text{HNC}})}{\tau} \right) \right)$$

$$\mathcal{L}_t = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp \left(\frac{\text{sim}(T_i, I_i)}{\tau} \right)}{\exp \left(\frac{\text{sim}(T_i, I_i)}{\tau} \right) + \sum_{j \neq i} \exp \left(\frac{\text{sim}(T_i, I_j)}{\tau} \right)}$$

$$\mathcal{L}_{\text{total}} = \frac{1}{2} (\mathcal{L}_i + \mathcal{L}_t)$$

Regularization Against Catastrophic Forgetting

1. L2 regularization

$$\text{Reg}_{L2} = \sum_i (\theta_i - \theta_i^{(CLIP)})^2$$

2. KL divergence

$$s_j = \frac{\exp(s_j/T)}{\sum_k \exp(s_k/T)}, \quad t_j = \frac{\exp(t_j/T)}{\sum_k \exp(t_k/T)}, \quad D_{\text{KL}}(t \parallel s) = \sum_{j=1}^{2B} t_j \log \frac{t_j}{s_j},$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{contrastive}} + \lambda \cdot \text{Reg}$$

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Experiments: Datasets

- Training: 100k samples from HNC training dataset.
- Validation: 9k samples from HNC validation dataset.
- Testing:
 - HNC human-annotated set.
 - COCO-derived relational caption pairs.

Test Dataset

- Human-annotated
- No hard mapping for positive and negative captions

EXAMPLE:

"n6309.jpg":

```
{  
  "0": {"caption": "The people are behind the street", "type": "relation", "label": 0, "replaced": "street, car"},  
  "1": {"caption": "The people are behind the car", "type": "relation", "label": 1, "replaced": null},  
  "2": {"caption": "The horses are behind the trees", "type": "relation", "label": 0, "replaced": "behind, in front of"},  
  "3": {"caption": "The horses are in front of the trees", "type": "relation", "label": 1, "replaced": null},  
  "4": {"caption": "The car is riding the horses.", "type": "relation", "label": 0, "replaced": "men, car"},  
  "5": {"caption": "The men are riding the horses.", "type": "relation", "label": 1, "replaced": null}  
}
```

- Problem: Not always following {0, 1, 0, 1, 0, 1... }

HNC Test Dataset Pairing Strategy

- Grouped into consecutive pairs (i and $i+1$)
- If (0, 0) or (1, 1) then skip this pair
- If 'type' is different, then skip this pair
- Select label=1 as positive in each pair
- Humanly check

Test dataset: 950 samples-->After filtering : **872 samples**

Training parameters

Initialized from the publicly released github CLIP checkpoint.

To control which parts of CLIP are updated during fine-tuning, several parameter selection modes are supported.

Mode	Text Encoder	Text Projection	Visual Encoder	Vision Projection
text_encoder	✓	✓	—	—
vision_encoder	—	—	✓	✓
full_encoder	✓	✓	✓	✓
last_encoder	last block	—	last block	—

Evaluation Metrics

Positive Score:

- Sim(Positive capture, Image)

HNC Score:

- Sim(Hard negative caption, Image)

Margin:

- Sim(Positive capture, Image) - Sim(Hard negative caption, Image)

Random Negative Score

- Sim(Random negative, Image)

Threshold-based accuracy:

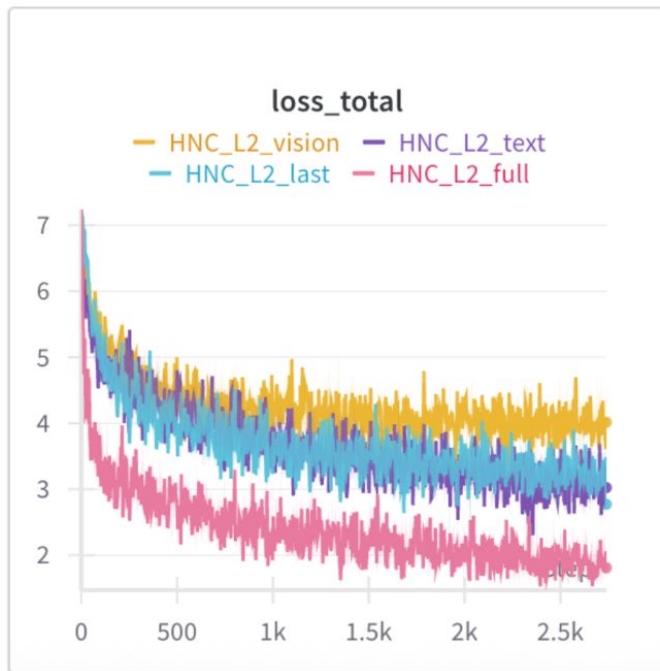
$$\frac{s_{\text{pos}}}{s_{\text{neg}}} > \tau$$

- To quantify this contrastive separation
- E.g. Acc@1 means: Sim(Positive, Image)/Sim(HNC, Image)>1

Outline

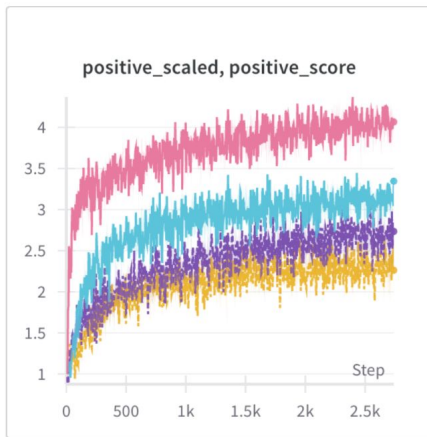
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Behavior Across Different Training Parameter Modes

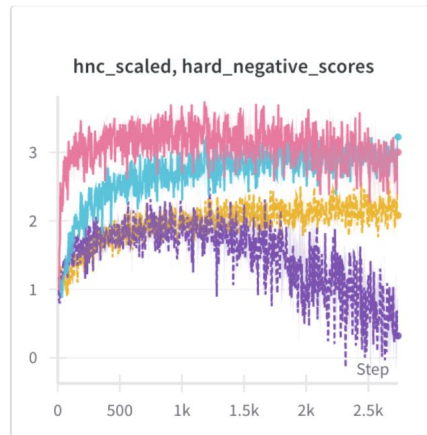


- 20 epoch
- Loss convergence: All configurations are able to fit the training data efficiently in the early stages

Behavior Across Different Training Parameter Modes

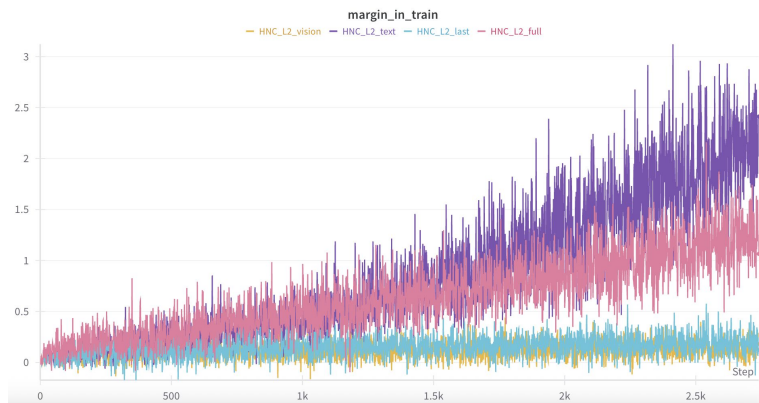


- Positive score get improved
- **Full encoder** configuration showing the largest increase and highest final score



- **Vision encoder** and **last encoder**: increase – level off
 - Models struggle to further suppress semantically similar distractors after an early stage of learning
- **Full encoder** and **text encoder**: increase - decline
 - Growing ability to suppress semantically incorrect captions
- **Text encoder** plays a crucial role, only **visual encoder** provides limited improvement

Behavior Across Different Training Parameter Modes



- Text encoder and full encoder : increase steadily
 - Indicating growing contrastive separation between positive and hard negative captions.
- Vision encoder and last encoder: very limited growth
- Text encoder plays a crucial role

Test scores of different training parameter modes

Each model is selected based on **validation** performance.

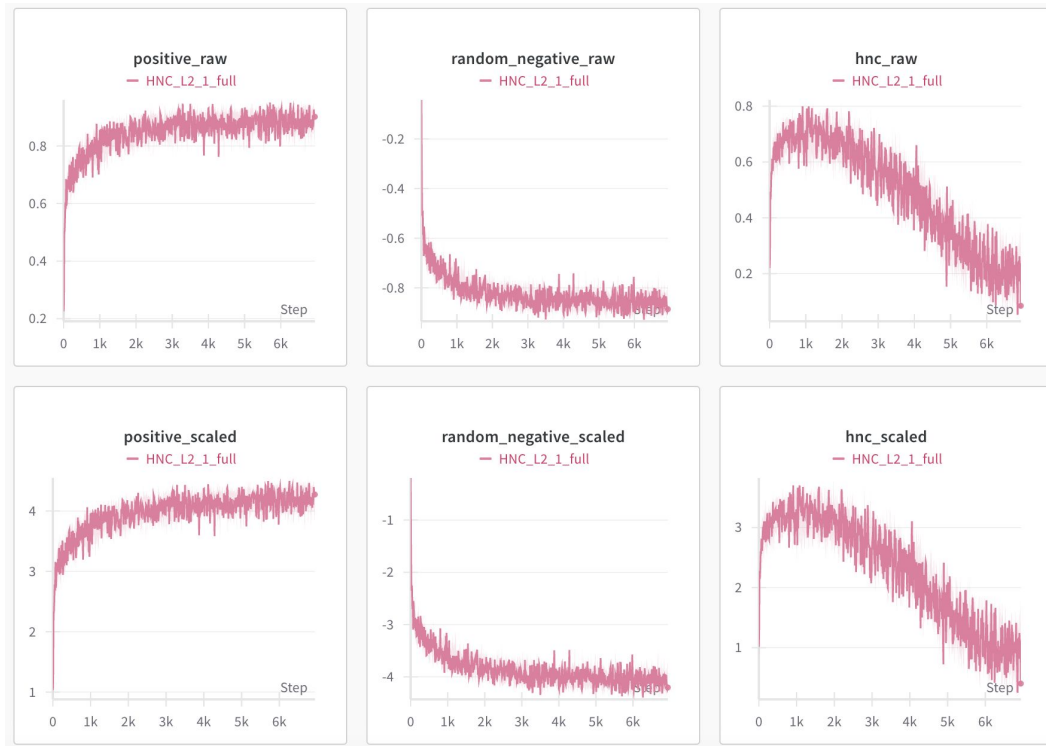
Acc@1 : positive similarity score is greater than or equal to the negative similarity score

In model name: 'L2' means using L2 regularization, 'text' means only the text encoder is updated, as such

Model	Avg_Pos	Avg_Neg	Margin	Rand_Neg	Acc@1
ViT-B/32 (Base)	0.2478	0.2457	0.0021	0.1769	0.5459
HNC_L2_text	0.2448	0.2396	0.0052	-0.0020	0.4587
HNC_L2_vision	0.3552	0.3465	0.0087	0.0035	0.5103
HNC_L2_last	0.4639	0.4515	0.0124	0.0277	0.5264
HNC_L2_full	0.5799	0.5636	0.0163	-0.0028	0.5287

Full encoder performance best: largest positive similarity, largest margin, and lowest random negative score.

Training progression for the full encoder model



During training:

Positive score -> Increased
Random negative score -> Drop
Hard negative score -> Rise then decline

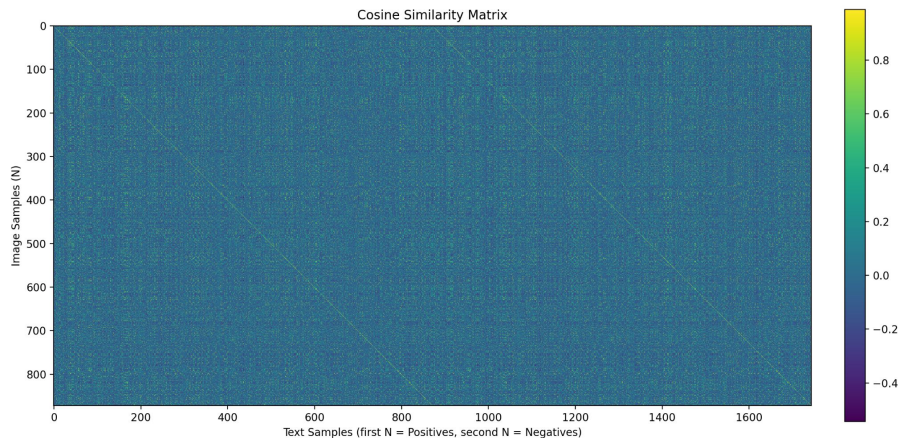
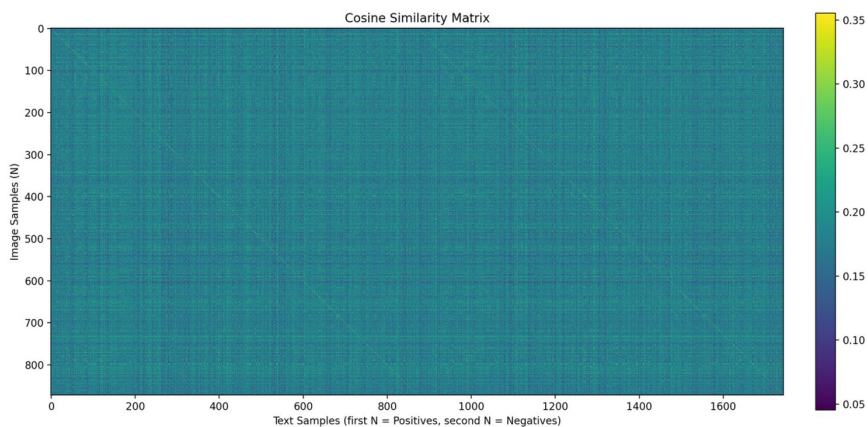
→ Model separates positive and random negative captions then gradually learns to separate positive captions and HNC.

To observe the trend, here running **50 epoch**, 137 step/epoch, $137 \times 50 = 6850$ steps

Cosine Similarity Visualization

Cosine similarity score of image and (positive + hard negative caption)

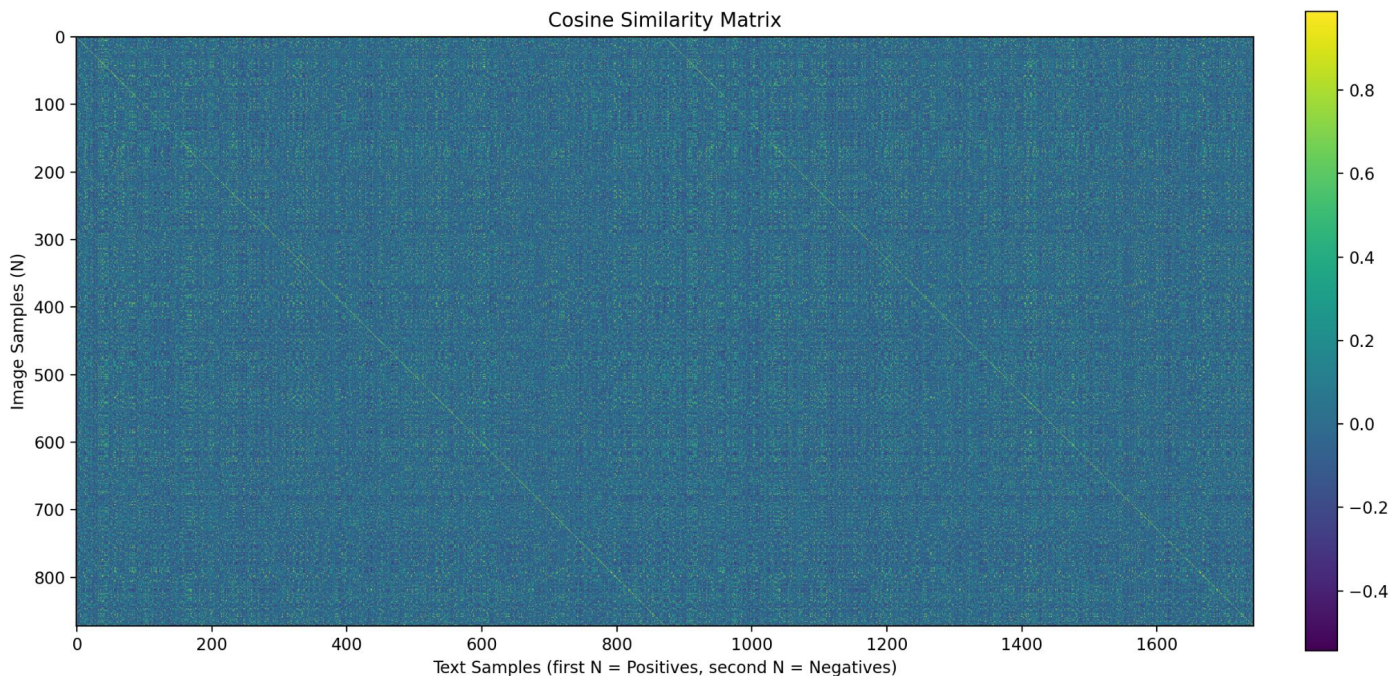
Base model (left): cosine similarity scores are relatively low and lack a clear diagonal structure
HNC fine tuned model(right): exhibits two diagonal structures



Cosine Similarity Visualization

Left diagonal corresponds to image–positive caption pairs, while the right diagonal represents image–hard negative caption pairs.

Because the hard negative captions still contain a lot of image information. So HNC scores are higher than random negative scores.



Comparison of Regularization Strategies: L2 vs. KL

Metric	Base	L2_text	L2_vision	L2_last	L2_full	KL_full
Avg_Pos	0.2478	0.2448	0.3552	0.4639	0.5799	0.6023
Avg_Neg	0.2457	0.2396	0.3465	0.4515	0.5636	0.5779
Margin	0.0021	0.0052	0.0087	0.0124	0.0163	0.0244
Rand_Neg	0.1769	-0.0020	0.0035	0.0277	-0.0028	0.0093
Acc@1	0.5459	0.4587	0.5103	0.5264	0.5287	0.5631
Acc@1.1	0.1227	0.2695	0.2489	0.2592	0.2993	0.2672
Acc@1.2	0.0470	0.1927	0.1755	0.1778	0.2167	0.1927
Acc@1.5	0.0000	0.1261	0.0860	0.0963	0.1227	0.1055
Acc@2	0.0000	0.0849	0.0482	0.0550	0.0711	0.0608
Acc@3	0.0000	0.0493	0.0241	0.0310	0.0333	0.0252

L2: Under stricter semantic threshold(Acc@1.1 and Acc@1.2) performance better

KL divergence: Higher positive scores, higher margin, higher Acc@1 score

- Stronger ability to identify the correct caption among each pair
- Better to prevent catastrophic forgetting in this case

Performance on COCO-Derived Test Set

COCO test dataset: random select 1000 samples

- Average positive score increased
- Margin increased
- Acc@1 decreased

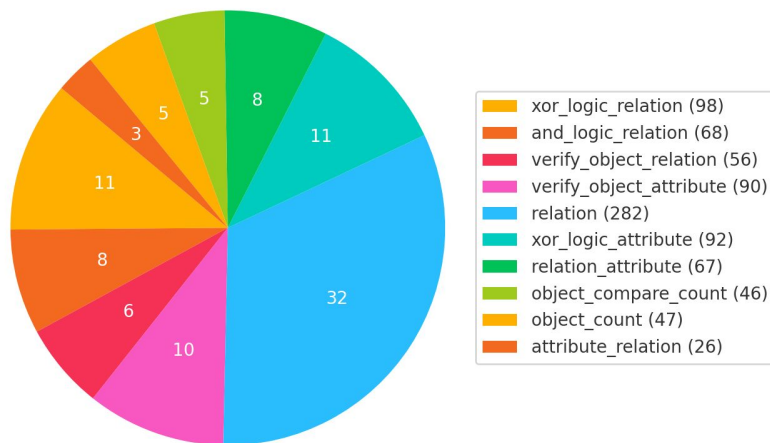
→ Strong generalization beyond the training domain

Metric	Base	L2_full	KL_full
Avg_Pos	0.3043	0.7198	0.7407
Avg_Neg	0.2973	0.7020	0.7255
Margin	0.0071	0.0178	0.0151
Avg_Rand_Neg	0.1510	0.0265	0.0231
Acc@1	0.7160	0.6180	0.6290
Acc@1.1	0.0750	0.1740	0.1170
Acc@1.2	0.0090	0.0930	0.0630
Acc@1.5	0.0000	0.0320	0.0160
Acc@2	0.0000	0.0150	0.0080
Acc@3	0.0000	0.0060	0.0000

Limitations

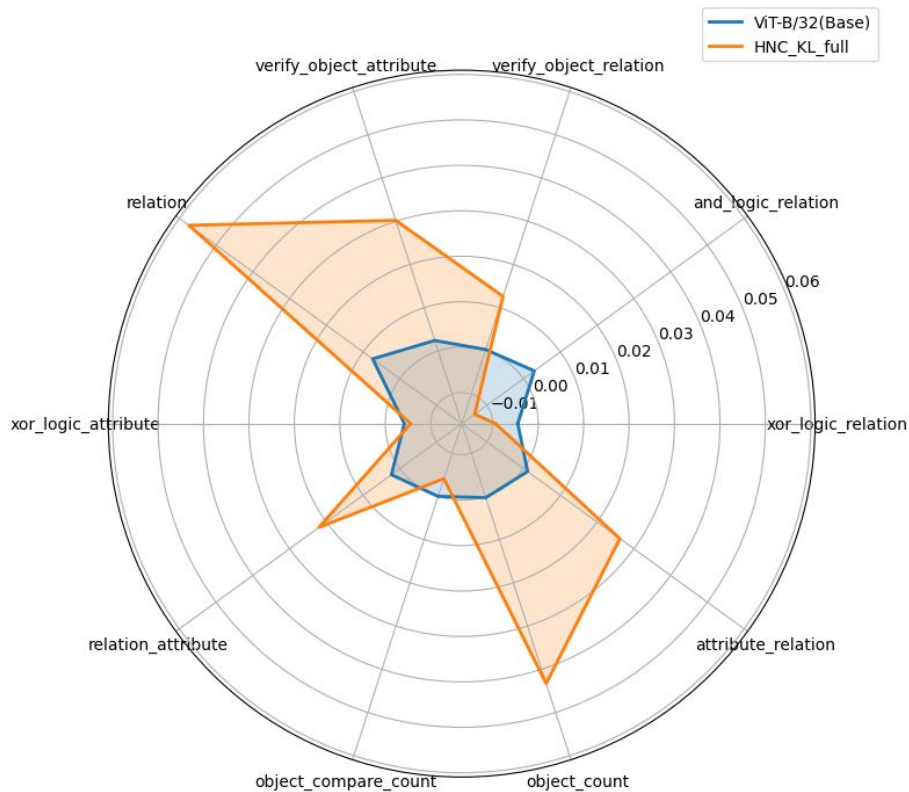
Still struggles with complex logic and coreference.

Performance by Semantic Type:



Type	Samples	ViT-B/32(Base)	HNC_KL_full	Improvement
ALL	872	0.0021	0.0244	0.0223
relation	282	0.0074	0.0574	0.0500
xor_logic_relation	98	-0.0045	-0.0094	-0.0049
xor_logic_attribute	92	-0.0042	-0.0056	-0.0014
verify_object_attribute	90	0.0024	0.0302	0.0278
and_logic_relation	68	0.0029	-0.0133	-0.0162
relation_attribute	67	0.0023	0.0220	0.0197
verify_object_relation	56	0.0003	0.0126	0.0123
object_count	47	0.0003	0.0433	0.0430
object_compare_count	46	0.0000	-0.0041	-0.0041
attribute_relation	26	0.0011	0.0262	0.0251

Comparing margin scores by type



Good at:

Spatial and object-level semantics
through contrastive fine-tuning

Bad at:

Logic-intensive categories

- 'and_logic_relation'
- 'xor_logic_relation'
- 'object_compare_count'
- 'xor_logic_attribute'

Examples of Challenging Type

Type - **and_logic_relation**

Pos: There are **both** a lamp **behind** the women **and** a laptop in front of the window.

Neg: There are **both** a lamp **in front of** the women **and** a laptop in front of the window.

Type - **xor_logic_relation**

Pos: The laptop is **either** behind the women **or** the sofa.

Neg: The laptop is **either** behind the women **or** the foot.

Type - **object_compare_count**

Pos: There are **as many** remote controls **as** people.

Neg: There are **fewer** remote controls **than** people.

Type - **xor_logic_attribute**

Pos: There is **either** a red carpet **or** green curtains.

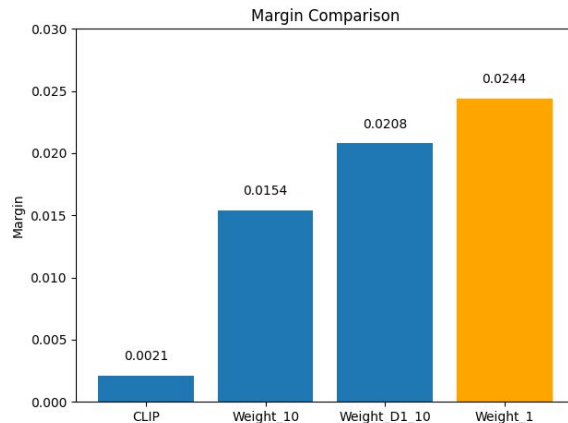
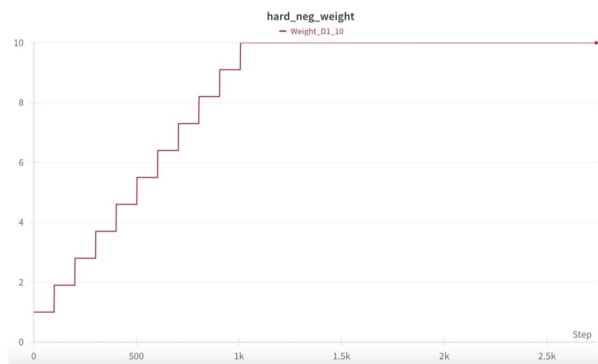
Neg: There is **either** a red carpet **or** purple curtains.



Limitations

Overweighting hard negatives may hurt stability

- Larger HNC weights do not always lead to better performance.
- Using dynamic weight (small at first, increasing the weight during training), better than fixed large weight, but still performance bad.



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Visual Interpretability

Based on a second-order attributions method for CLIP-based models (Tilli et al.)

- Decomposes the similarity score into fine-grained contributions from image-text feature pairs
- Captures cross-modal interactions between specific words and regions

Pos: The trees are behind the fence

- text-to-image

Original Image



Base CLIP
score=0.0430



Fine-tuned CLIP
score=0.0504



Thanks to Dr. Pascal Tilli and their team.

<https://arxiv.org/abs/2408.14153>

Visual Interpretability

Pos: The trees are behind the fence

- Token-level image-to-text attribution visualization

Base CLIP

the trees are behind the fence

Fine-tuned CLIP

the trees are behind the fence

Visual Interpretability

Pos: The **trees** are behind the fence.

Base CLIP: "trees"



Fine-tuned CLIP: "trees"



Visual Interpretability

Pos: The trees are behind the **fence**.

Base CLIP: "fence"



Fine-tuned CLIP: "fence"



→ Object detection get improved. E.g. trees, fence

Visual Interpretability

Pos: The trees are **behind** the fence.

Base CLIP: "behind"



Fine-tuned CLIP: "behind"



→ For relational words (e.g. behind) can also focus on the right place in image.

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Results Summary

- Relational understanding improved.
- Text encoder plays a central role.
- Full encoder tuning achieves the best overall performance.
- KL regularization yields highest margin, can better deal with catastrophic forgetting in this case.
- L2 performance better at strict thresholds.
- Good out-of-domain generalization (On a COCO-derived test set).
- Visual interpretability improved.

Thank you

✨ Special Thanks ✨

🌸 Dr. Pascal Tilli 🌸

🌻 Prof. Dr. Ngoc Thang Vu 🌻

🌿 Weekly Thesis Meeting 🌿